

Лабораторная работа №16

Настройка VPN

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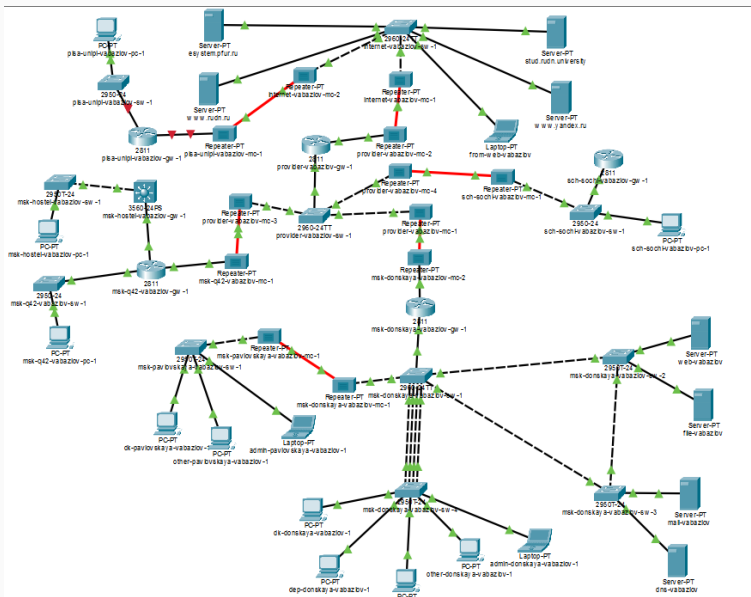
22 мая 2026

Российский университет дружбы народов, Москва, Россия

Цель работы

Получить навыки настройки VPN-туннеля через незащищённое Интернет-соединение.

Размещение оборудования



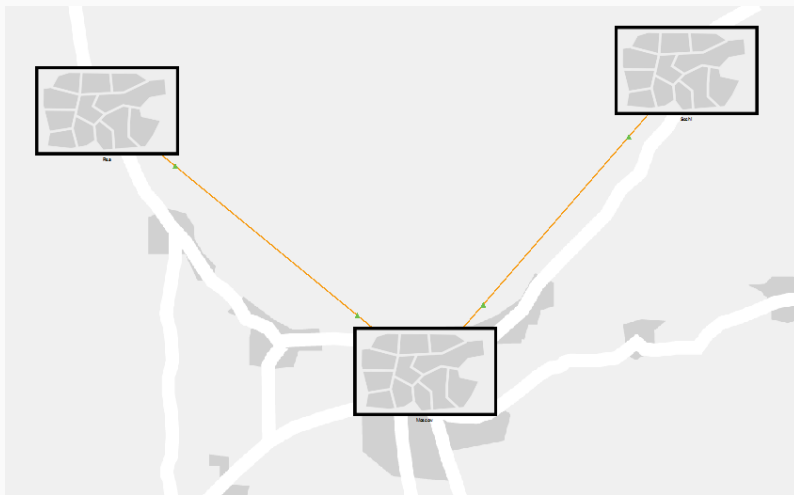
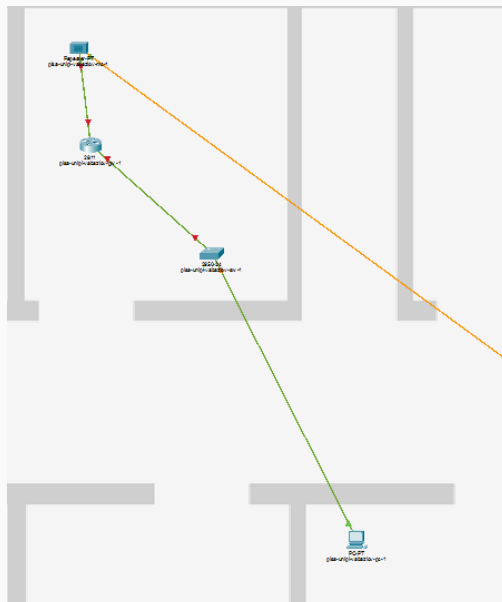


Рис. 2: Создание города Пиза в физической рабочей области



Первоначальная настройка сети

Настройка коммутатора

```
pisa-unipi-vabazlov-sw-1#
pisa-unipi-vabazlov-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
pisa-unipi-vabazlov-sw-1(config)#int f0/24
pisa-unipi-vabazlov-sw-1(config-if)#sw mode trunk
pisa-unipi-vabazlov-sw-1(config-if)#int f0/1
pisa-unipi-vabazlov-sw-1(config-if)#sw mode acc
pisa-unipi-vabazlov-sw-1(config-if)#sw acc vlan 401
% Access VLAN does not exist. Creating vlan 401
pisa-unipi-vabazlov-sw-1(config-if)#vlan 401
pisa-unipi-vabazlov-sw-1(config-vlan)#name unipi-main
pisa-unipi-vabazlov-sw-1(config-vlan)#int vlan 401
pisa-unipi-vabazlov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed state to up

pisa-unipi-vabazlov-sw-1(config-if)#no sh
pisa-unipi-vabazlov-sw-1(config-if)#exit
pisa-unipi-vabazlov-sw-1(config)#exit
pisa-unipi-vabazlov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

pisa-unipi-vabazlov-sw-1#wr mem
Building configuration...
[OK]
pisa-unipi-vabazlov-sw-1#
```

Рис. 4: Настройка VLAN 401 и портов коммутатора

Настройка маршрутизатора

```
pisa-unipi-vabazlov-gw-1#
pisa-unipi-vabazlov-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
pisa-unipi-vabazlov-gw-1(config)#
pisa-unipi-vabazlov-gw-1(config)#int f0/0
pisa-unipi-vabazlov-gw-1(config-if)#no sh

pisa-unipi-vabazlov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

pisa-unipi-vabazlov-gw-1(config-if)#int f0/0.401
pisa-unipi-vabazlov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.401, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up

pisa-unipi-vabazlov-gw-1(config-subif)#enc dot1q 401
pisa-unipi-vabazlov-gw-1(config-subif)#ip addr 10.131.0.1 255.255.255.0
pisa-unipi-vabazlov-gw-1(config-subif)#descr unipi-main
pisa-unipi-vabazlov-gw-1(config-subif)#int f0/1
pisa-unipi-vabazlov-gw-1(config-if)#no sh

pisa-unipi-vabazlov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

pisa-unipi-vabazlov-gw-1(config-if)#ip addr 192.0.2.20 255.255.255.0
pisa-unipi-vabazlov-gw-1(config-if)#descr internet
pisa-unipi-vabazlov-gw-1(config-if)#exit
pisa-unipi-vabazlov-gw-1(config)#ip route 0.0.0.0 0.0.0.0 192.0.2.1
pisa-unipi-vabazlov-gw-1(config)#
```

Рис. 5: Настройка подынтерфейса VLAN 401 и внешнего интерфейса

Настройка VPN GRE

```
pisa-unipi-vabazlov-gw-1(config)#
pisa-unipi-vabazlov-gw-1(config)#int tunnel 0

pisa-unipi-vabazlov-gw-1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

pisa-unipi-vabazlov-gw-1(config-if)#ip addr 10.128.255.254 255.255.255.252
pisa-unipi-vabazlov-gw-1(config-if)#tun sou f0/1
pisa-unipi-vabazlov-gw-1(config-if)#tun des 198.51.100.2
pisa-unipi-vabazlov-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

pisa-unipi-vabazlov-gw-1(config-if)#exit
pisa-unipi-vabazlov-gw-1(config)#int loopback 0

pisa-unipi-vabazlov-gw-1(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

pisa-unipi-vabazlov-gw-1(config-if)#ip addr 10.128.254.5 255.255.255.255
pisa-unipi-vabazlov-gw-1(config-if)#exit
pisa-unipi-vabazlov-gw-1(config)#
pisa-unipi-vabazlov-gw-1(config)#ip route 10.128.254.1 255.255.255.255 10.128.255.253
pisa-unipi-vabazlov-gw-1(config)#
pisa-unipi-vabazlov-gw-1(config)#router ospf 1
pisa-unipi-vabazlov-gw-1(config-router)#router-id 10.128.254.5
pisa-unipi-vabazlov-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
pisa-unipi-vabazlov-gw-1(config-router)#exit
pisa-unipi-vabazlov-gw-1(config)#exit
pisa-unipi-vabazlov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

pisa-unipi-vabazlov-gw-1#wr mem
Building configuration...
[OK]
pisa-unipi-vabazlov-gw-1#
```

GRE-туннель на удалённой стороне

```
nsk-donskaya-vabazlov-gw-1>en
Password:
nsk-donskaya-vabazlov-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
nsk-donskaya-vabazlov-gw-1(config)#
nsk-donskaya-vabazlov-gw-1(config)#int tunnel 0

nsk-donskaya-vabazlov-gw-1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

nsk-donskaya-vabazlov-gw-1(config-if)#ip addr 10.128.255.253 255.255.255.252
nsk-donskaya-vabazlov-gw-1(config-if)#tun sou f0/1.4
nsk-donskaya-vabazlov-gw-1(config-if)#tun dest 192.0.2.20
nsk-donskaya-vabazlov-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

nsk-donskaya-vabazlov-gw-1(config-if)#
nsk-donskaya-vabazlov-gw-1(config-if)#exit
nsk-donskaya-vabazlov-gw-1(config)#
nsk-donskaya-vabazlov-gw-1(config)#int loopback 0

nsk-donskaya-vabazlov-gw-1(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

04:55:35: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.5 on Tunnel0 from LOADING to FULL, Loading Done

nsk-donskaya-vabazlov-gw-1(config-if)#ip addr 10.128.254.1 255.255.255.255
nsk-donskaya-vabazlov-gw-1(config-if)#exit
nsk-donskaya-vabazlov-gw-1(config)#ip route 10.128.254.5 255.255.255.255 10.128.255.254
nsk-donskaya-vabazlov-gw-1(config)#exit
nsk-donskaya-vabazlov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

nsk-donskaya-vabazlov-gw-1#wr mem
Building configuration...
[OK]
nsk-donskaya-vabazlov-gw-1#
```

Настройка рабочей станции

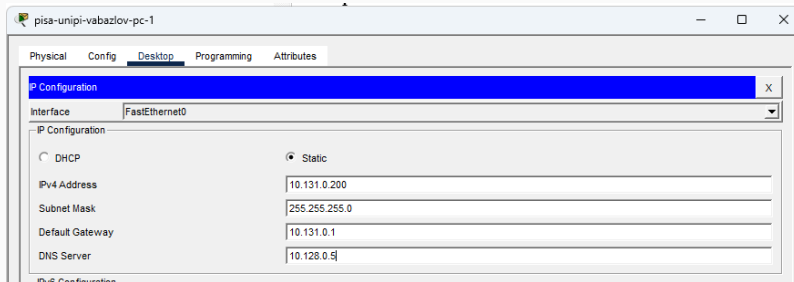


Рис. 8: Настройка IP-адреса рабочей станции Университета г. Пиза

Проверка доступности

Проверка с ноутбука администратора

```
admin-donskaya-vabazlov-1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 10.131.0.1

Pinging 10.131.0.1 with 32 bytes of data:

Reply from 10.131.0.1: bytes=32 time<1ms TTL=254
Reply from 10.131.0.1: bytes=32 time<1ms TTL=254
Reply from 10.131.0.1: bytes=32 time<1ms TTL=254
Reply from 10.131.0.1: bytes=32 time=2ms TTL=254

Ping statistics for 10.131.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 10.131.0.200

Pinging 10.131.0.200 with 32 bytes of data:

Request timed out.
Reply from 10.131.0.200: bytes=32 time=2ms TTL=126
Reply from 10.131.0.200: bytes=32 time=10ms TTL=126
Reply from 10.131.0.200: bytes=32 time<1ms TTL=126

Ping statistics for 10.131.0.200:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 4ms

C:\>tracert 10.131.0.200

Tracing route to 10.131.0.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.128.6.1
  2  0 ms    1 ms    0 ms    10.128.255.254
  3  0 ms    1 ms    0 ms    10.131.0.200

Trace complete.

C:\>
```

Проверка маршрутизации

Таблица маршрутизации сети «Донская»

```
nsk-donskaya-vabazlov-gw-1#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 29 subnets, 4 masks
C       10.128.0.0/24 is directly connected, FastEthernet0/0.3
L       10.128.0.1/32 is directly connected, FastEthernet0/0.3
C       10.128.1.0/24 is directly connected, FastEthernet0/0.2
L       10.128.1.1/32 is directly connected, FastEthernet0/0.2
C       10.128.3.0/24 is directly connected, FastEthernet0/0.101
L       10.128.3.1/32 is directly connected, FastEthernet0/0.101
C       10.128.4.0/24 is directly connected, FastEthernet0/0.102
L       10.128.4.1/32 is directly connected, FastEthernet0/0.102
C       10.128.5.0/24 is directly connected, FastEthernet0/0.103
L       10.128.5.1/32 is directly connected, FastEthernet0/0.103
C       10.128.6.0/24 is directly connected, FastEthernet0/0.104
L       10.128.6.1/32 is directly connected, FastEthernet0/0.104
C       10.128.254.1/32 is directly connected, Loopback0
S       10.128.254.5/32 [1/0] via 10.128.255.254
C       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.1/32 is directly connected, FastEthernet0/1.5
C       10.128.255.4/30 is directly connected, FastEthernet0/1.6
L       10.128.255.5/32 is directly connected, FastEthernet0/1.6
O       10.128.255.8/30 [110/2] via 10.128.255.2, 00:18:36, FastEthernet0/1.5
        [110/2] via 10.128.255.6, 00:18:36, FastEthernet0/1.6
C       10.128.255.252/30 is directly connected, Tunnel0
L       10.128.255.253/32 is directly connected, Tunnel0
S       10.129.0.0/16 [1/0] via 10.128.255.2
O       10.129.0.0/24 [110/2] via 10.128.255.2, 00:18:46, FastEthernet0/1.5
O       10.129.1.0/24 [110/2] via 10.128.255.2, 00:18:46, FastEthernet0/1.5
O       10.129.128.0/24 [110/3] via 10.128.255.2, 00:18:46, FastEthernet0/1.5
S       10.130.0.0/16 [1/0] via 10.128.255.6
O       10.130.0.0/24 [110/2] via 10.128.255.6, 00:18:46, FastEthernet0/1.6
O       10.130.1.0/24 [110/2] via 10.128.255.6, 00:18:46, FastEthernet0/1.6
O       10.131.0.0/24 [110/1001] via 10.128.255.254, 00:02:19, Tunnel0
    198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/28 is directly connected, FastEthernet0/1.4
L       198.51.100.2/32 is directly connected, FastEthernet0/1.4
S*      0.0.0.0/0 [1/0] via 198.51.100.1

nsk-donskaya-vabazlov-gw-1#
```

Таблица маршрутизации сети Пизы

```
pisa-unipi-vabazlov-gw-1#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.0.2.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 20 subnets, 3 masks
O       10.128.0.0/24 [110/1001] via 10.128.255.253, 00:02:50, Tunnel0
O       10.128.1.0/24 [110/1001] via 10.128.255.253, 00:02:50, Tunnel0
O       10.128.3.0/24 [110/1001] via 10.128.255.253, 00:02:50, Tunnel0
O       10.128.4.0/24 [110/1001] via 10.128.255.253, 00:02:50, Tunnel0
O       10.128.5.0/24 [110/1001] via 10.128.255.253, 00:02:50, Tunnel0
O       10.128.6.0/24 [110/1001] via 10.128.255.253, 00:02:50, Tunnel0
S       10.128.254.1/32 [1/0] via 10.128.255.253
C       10.128.254.5/32 is directly connected, Loopback0
O       10.128.255.0/30 [110/1001] via 10.128.255.253, 00:02:50, Tunnel0
O       10.128.255.4/30 [110/1001] via 10.128.255.253, 00:02:50, Tunnel0
O       10.128.255.8/30 [110/1002] via 10.128.255.253, 00:02:50, Tunnel0
C       10.128.255.252/30 is directly connected, Tunnel0
L       10.128.255.254/32 is directly connected, Tunnel0
O       10.129.0.0/24 [110/1002] via 10.128.255.253, 00:00:52, Tunnel0
O       10.129.1.0/24 [110/1002] via 10.128.255.253, 00:00:52, Tunnel0
O       10.129.128.0/24 [110/1003] via 10.128.255.253, 00:00:52, Tunnel0
O       10.130.0.0/24 [110/1002] via 10.128.255.253, 00:02:50, Tunnel0
O       10.130.1.0/24 [110/1002] via 10.128.255.253, 00:02:50, Tunnel0
C       10.131.0.0/24 is directly connected, FastEthernet0/0.401
L       10.131.0.1/32 is directly connected, FastEthernet0/0.401
    192.0.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.0.2.0/24 is directly connected, FastEthernet0/1
L       192.0.2.20/32 is directly connected, FastEthernet0/1
S*    0.0.0.0/0 [1/0] via 192.0.2.1
```

Итоги работы

В ходе лабораторной работы:

- размещено оборудование сети Университета г. Пиза;
- создана физическая структура проекта;
- настроена VLAN 401;
- настроены интерфейсы маршрутизатора;
- создан GRE-туннель между удалёнными площадками;
- настроен обмен маршрутами через OSPF;
- проверена доступность узлов сети Университета г. Пиза.

Получены практические навыки настройки VPN GRE, VLAN, маршрутизации и проверки связности удалённых сетей.